LLM-Trader: A Multimodal RAG Approach for Generating Trading Decisions using LLMs

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Abstract

This paper presents a hybrid architecture for trading Bitcoin that integrates time series data and financial news to make well-founded trading decisions. The system employs a retrieval-augmented generation (RAG) approach that extracts relevant price patterns and news headlines from historical 2022 data and aligns them with market conditions during the 2023 test period. The resulting context is fed into a language model (Qwen2.5 0.5B), which outputs a trading decision (BUY/SELL/HOLD) along with its rationale. During testing, the system achieved a total return of +185.3%, outperforming the Buy & Hold strategy (154.5%). It is demonstrated that enriching an LLM with historical multimodal context leads to interpretable and effective decisions in financial forecasting tasks.

1 Introduction

Cryptocurrency markets are characterized by high volatility and unstable dependencies, making the construction of robust models challenging. Traditional approaches based solely on time series analysis ignore the impact of external factors such as news and macroeconomic events.

On the other hand, simply adding current news to the LLM's context does not provide insight into the trend and quantitative characteristics of the time series.

This work introduces a **hybrid architecture** that combines modern retrieval techniques from documents with current language model architectures. The model uses a vector representation of the current market state and, through its search algorithm, finds similar historical situations to feed them into the language model for decision-making. This approach allows the model to reason by analogy based on historical context.

2 Related Work

TS2Vec [1] proposed hierarchical contrastive learning for universal time series embeddings. FinBERT [2] and FinRL [3] combine text and price data, but lack deep integration. RAG pipelines [4] enable the integration of retrieved context with generation. Methods for joint processing of text and time series are actively being studied in finance [5]. In this work, retrieval is applied prior to reasoning, thereby enhancing model interpretability.

Unlike the Chronos model [6], which employs a closed tokenizer and extensive training, the presented method does not require additional fine-tuning and works with an open and lightweight model, making it simpler and more reproducible.

It is also worth mentioning TS-RAG [7], which applies retrieval-augmented generation to time series tasks by extracting semantically similar windows and merging them with the query via a MoE module. In contrast to TS-RAG, the architecture presented here operates without additional training and demonstrates strong results solely based on the tuning of semantic search and generation hyperparameters.

3 Methodology

3.1 Data

- **Price Data**: BTC/USD, 1-minute interval for 2022. Sourced from the *Bitcoin Price Dataset (2017–2023)* [10] on Kaggle. The data includes fields such as Timestamp, Open, High, Low, Close, Volume, etc.
- **Indicators**: Computed from price data: Close, SMA_{10}, EMA_{10}, RSI, MACD (see Appendix).

- **Windows**: A sliding window of length 64 with a stride of 1 minute.
- **News**: Headlines with sentiment scores, sorted by date. The dataset used is *Sentiment Analysis of Bitcoin News (2021–2024)* [11] from Kaggle. A fine-tuned transformer model is applied for sentiment analysis.

3.2 Retrieval and Context Construction Algorithm

For each input window and news item from 2023, a multimodal context is constructed as follows:

- **Retrieval of Similar Price Windows:**
  - Index the vector representations of indicator windows (SMA, EMA, RSI, etc.) from 2022 using FAISS for efficient search.
  - Retrieve the top-10 most similar price windows from 2022 based on the L2 distance between indicator embeddings.
- **Retrieval of Relevant News:**
  - Index the 2022 news texts using BM25 for full-text search.
  - For each daily news item in 2023, extract the top-5 most relevant 2022 news headlines based on cosine similarity of text embeddings (using the all-MiniLM-L6-v2 model as the encoder; embedding dimension: 384).

3.3 Context-Augmented Retrieval Algorithm

Input: Time window W_{2023} (indicators + news)
Initialization: $\mathcal{C} \leftarrow \emptyset$
1. Price Pattern Retrieval:
- Compute L2 distances: $\|W^{\text{ind}}_{2023} - W^{\text{ind}}_{2022}\|_2$
- Select $\mathcal{P}_{\text{top10}} \leftarrow$ Top-10 closest windows
- $\mathcal{C} \leftarrow \mathcal{C} \cup \mathcal{P}_{\text{top10}}$

2. Semantic News Retrieval:
For each news item $n_i \in W^{\text{news}}_{2023}$:
- Compute cosine similarity: $\text{sim}(n_i, N_{2022}) = \frac{n_i \cdot n_j}{\|n_i\| \|n_j\|}$
- Select $\mathcal{N}_{\text{top5}} \leftarrow$ Top-5 with $\Delta t \leq 3$ days
- $\mathcal{C} \leftarrow \mathcal{C} \cup \mathcal{N}_{\text{top5}}$

3. Aggregation by Timestamp:
$\mathcal{C}_{\text{final}} = \text{AggregateByDate}(\mathcal{C})$
Return: $\mathcal{C}_{\text{final}}$

Table 1: Algorithm for constructing the final multimodal context based on time series and news data

3.4 Visualization of the Retrieval Space

t-SNE Interpretation: Figure 1 shows partial overlap of clusters, which supports the hypothesis of an interrelation between price movements and the news backdrop.

3.5 LLM Reasoning

For generating the trading decision, a compact language model **Qwen2.5-0.5B** [8, 9] is used. This model demonstrates a high capability for logical reasoning and flexibility by utilizing a structured output approach. Unlike other models designed for similar tasks, the chosen model is available under an open license.

- **Input**: Current technical indicators + retrieved price patterns + relevant news.
- **Output**: Structured JSON with the recommendation and explanation:

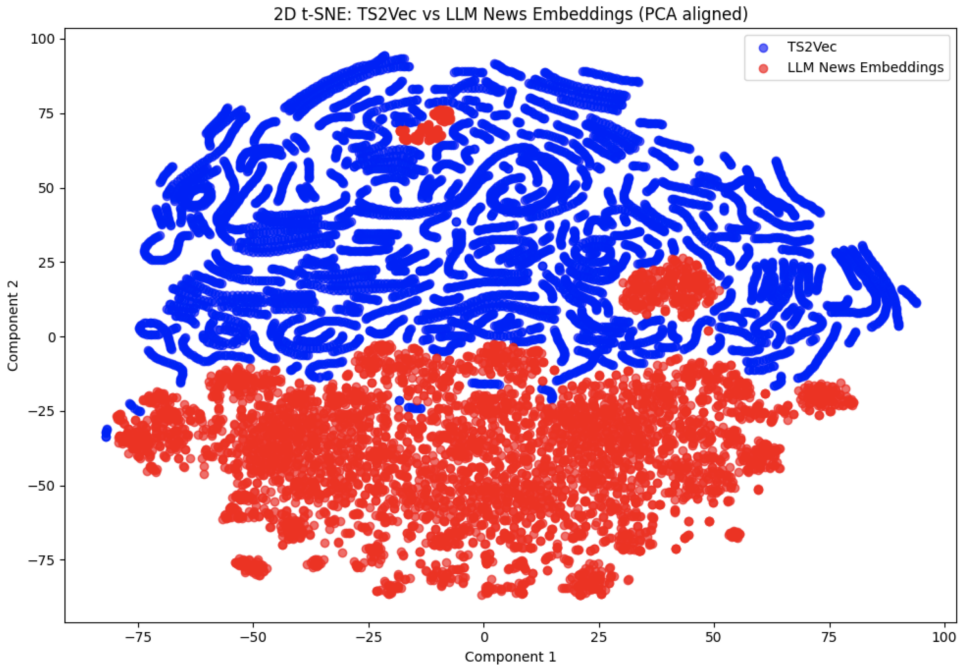


Figure 1: t-SNE visualization of news embeddings (red) and time series embeddings (blue). Shown is a 2D projection of the high-dimensional feature space. Clustering reflects semantic similarity: financial news group together with their corresponding price patterns.

```
{
  "recommendation": "BUY",
  "reasoning": {
    "price_pattern": { "RSI": 54.79, "MACD": 10.42 },
    "news_sentiment": {
      "average_score": -0.22,
      "interpretation": "Despite negative sentiment, the indicators show growth"
    }
  }
}
```

4 Experiments

Comparison of Strategies:



Figure 2: Trading results (news only)



Figure 3: Trading results (news + time series)

As seen in Table 2 and Figures 2 and 3, combining news with historical price windows delivers the highest return. However, the model using only news context exhibits more stable behavior (indicated by a higher Sharpe Ratio). Key observations:

- The time series strategy yields a 2.5x higher return compared to using "news only."

Strategy	Total Return	Annualized Return	Max Drawdown	Sharpe Ratio
LLM (price + news)	**185.29%**	**186.94%**	-20.51%	0.59
LLM (news only)	74.57%	75.11%	-20.03%	**1.49**
Buy & Hold	154.5%	—	-20.00%	1.00

Table 2: Backtest results of various strategies on 2023 data (initial capital $10,000)

- Both LLM strategies show similar maximum drawdowns (approximately 20%).
- The news-only model has an unusually high Sharpe Ratio (1.49 vs. 0.59 for the combined approach).

References

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Appendix: Indicator Formulas and Prompt

Technical Indicator Formulas

- **SMA$_n$ (Simple Moving Average)**:

$$\mathrm{SMA}_n(t) = \frac{1}{n}\sum_{i=0}^{n-1} P_{t-i}$$

  where: n is the averaging period, P_{t-i} is the price at time $t-i$.

- **EMA$_n$ (Exponential Moving Average)**:

$$\mathrm{EMA}_n(t) = \alpha \cdot P_t + (1-\alpha)\cdot \mathrm{EMA}_n(t-1)$$

  where $\alpha = \frac{2}{n+1}$.

- **RSI (Relative Strength Index)**:

$$\text{RSI}(t) = 100 - \left(\frac{100}{1 + RS}\right), \quad RS = \frac{\text{AvgGain}}{\text{AvgLoss}}$$

- **MACD**:

$$\text{MACD}(t) = \text{EMA}_{12}(t) - \text{EMA}_{26}(t)$$

Example Prompt for Generating Trading Decisions

```
You are a financial trading assistant AI.

Today's news (2023):
"Fed announces pause in rate hikes amid cooling inflation signals"

Relevant historical context:
- News (2022): "Inflation peaks at 9.1% as Fed raises rates by 75bps"
  (Sentiment: Negative, Score: -0.78)
- Price window (2022):
  Close: 4021.50, SMA_10: 3988.20, RSI: 62.4, MACD: 12.3

Technical indicators (current):
- RSI: 58.2 (neutral)
- MACD: 8.4 (bullish crossover)
- Price above SMA_50

Provide JSON response with trading action:
{"recommendation": "BUY|SELL|HOLD"}
```